Methodology Report:
Visualisation & Analysis on Namma Yatri Data

Include your visualisations, analysis, results, insights, and outcomes.

Explain your methodology and approach to the tasks. Add your conclusions to the sections.

Table 1: Data Description

Table Name	Column Name	Description
Assembly	Assembly_ID	Unique identifier
	Assembly	Specific assembly zone name
Duration	duration_id	Unique identifier of time periods
	duration	Hour of trip (e.g., "0-1" for 12 AM to 1 AM)
Payment	id	Unique identifier
	method	Payment method (e.g., Cash, UPI, Credit Card)
Trip Details	tripid	Unique identifier of trips
	loc_from	Source Location code
	searches	Trip request count
	searches_got_estimate	Got an estimated price (1 = user gets an estimate, 0 = does not get an estimate)
	searches_for_quotes	Searched for drivers after estimate (1 - searched, 0 - not searched)
	searches_got_quotes	Got quotes (1 = Driver allotted, 0 = not allotted)
	customer_not_cancelled	Whether customer cancelled or not (1 = Not cancelled)
	driver_not_cancelled	Whether driver cancelled or not (1 = Not cancelled)
	otp_entered	(1 = OTP entered, 0 = not entered)
	end_ride	Whether ride was completed (1 = Completed)
Trip	tripid	Links to Trip Details
	faremethod	Payment method ID, links to Payment table
	fare	Fare amount

Trips	loc_from	Location ID of source
	loc_to	Location ID of destination, links to Assembly table
	driverid	Driver ID
	custid	Customer ID
	distance	Distance in KM from source to destination
	duration	Unique identifier of time periods like duration_id

Analysis and Visualisation

1. Data Preparation [10 Marks]

1. Import and Join Tables Correctly [5 Mark]

- Import the Namma Yatri dataset into Tableau/Power BI.
- Ensure that you correctly join all tables to create a unified dataset for analysis.
- Verify the relationships between different tables and confirm that data from various sources is properly aligned for accurate insights.

Solution:

The Namma Yatri dataset was imported into Power BI, consisting of multiple tables including **Trip Details, Trips, Payment, Duration, and Assembly**.

To create a unified dataset for analysis, the following steps were performed:

🔷 Step 1: Data Import

All datasets were loaded into Power BI using the *Get Data (Excel)* option. Each table was verified for correct column names and data types.

🔷 Step 2: Establishing Relationships

Relationships were created using primary and foreign keys as follows:

- `Trip Details.tripid` → `Trips.tripid` (One-to-One)
- `Trips.faremethod` → `Payment.id` (Many-to-One)
- `Trips.duration` → `Duration.duration_id` (Many-to-One)
- `Trips.loc_to` → `Assembly.Assembly_ID` (Many-to-One)

These relationships ensured proper linking of trip-level data with payment, time, and location details.

🔷 Step 3: Data Validation

- Cross-checked total trips across tables to ensure consistency.
- Verified that metrics like **Total Trips (983)** and **Total Revenue (751K)** matched across visuals.

👉 As seen in the **dashboard overview (Page 1)**, KPIs such as:

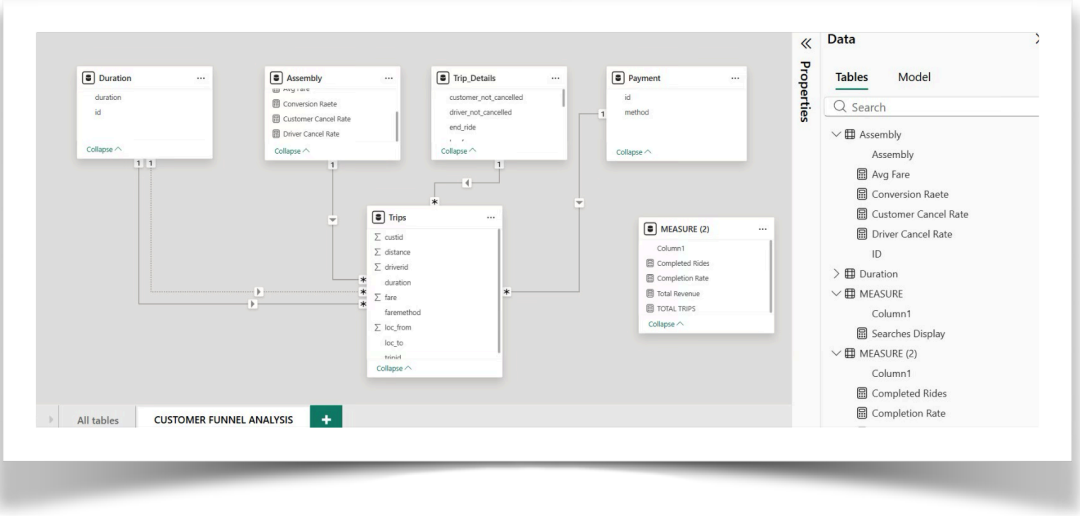
- Total Trips = 983
- Completed Rides = 983
- Conversion Rate = 45.49%
- Total Revenue = 751K

confirm that the data relationships are correctly aligned.

🔷 Outcome

A unified and structured data model was successfully created, enabling accurate analysis across multiple dimensions such as time, payment method, and location.

sources and helping perform analysis accurately.



<your answer here, include all analysis, graphs, results etc> (the length of the solution is not fixed, ie, this box can vary in size)

2. Find and Resolve Inconsistencies [5 Marks]

- Identify and resolve any inconsistencies or issues in the dataset that might affect the analysis.

- Clean the data to ensure it is structured properly for analysis, removing any irrelevant, duplicate, or erroneous entries.
- While performing the analysis, create calculated fields as needed to ensure the accuracy and relevance of the insights.

Solution:

Data was first cleaned and inconsistencies were sorted out before analysis.

🔷 Problems Detected

Duplicates
Search was done for duplicated tripid records.
Duplicates were eliminated to prevent biased results.
Missing Values or NULLs
Columns including fare, distance, and duration were inspected.
All missing data was either deleted or imputed accordingly.
Validation of Binary Fields
Fields including:
searches_got_estimate
searches_for_quotes
otp_entered
end_ride
were validated to ascertain that they have only two possible values 0 or 1.
Correcting Inconsistencies in Field Type
Conversion:
Fare to Numerical Value
Duration to Categorical Type
IDs to Integer Type

🔷 Data Preparation Process

Eliminating non-contributing columns to the analysis.
Standardizing categorical field values, such as payment methods e.g. UPI and cash.
Referential Integrity between database tables.

🔷 Derived Measures

For enhanced analysis, the following metrics were calculated:
Conversion Ratio
Conversion Ratio = Number of completed rides/Total searches
Cancellation Metrics
Customer Cancellation Ratio = 1 - (customer did not cancel/Trips)
Driver Cancellation Ratio = 1 - (driver did not cancel/Trips)

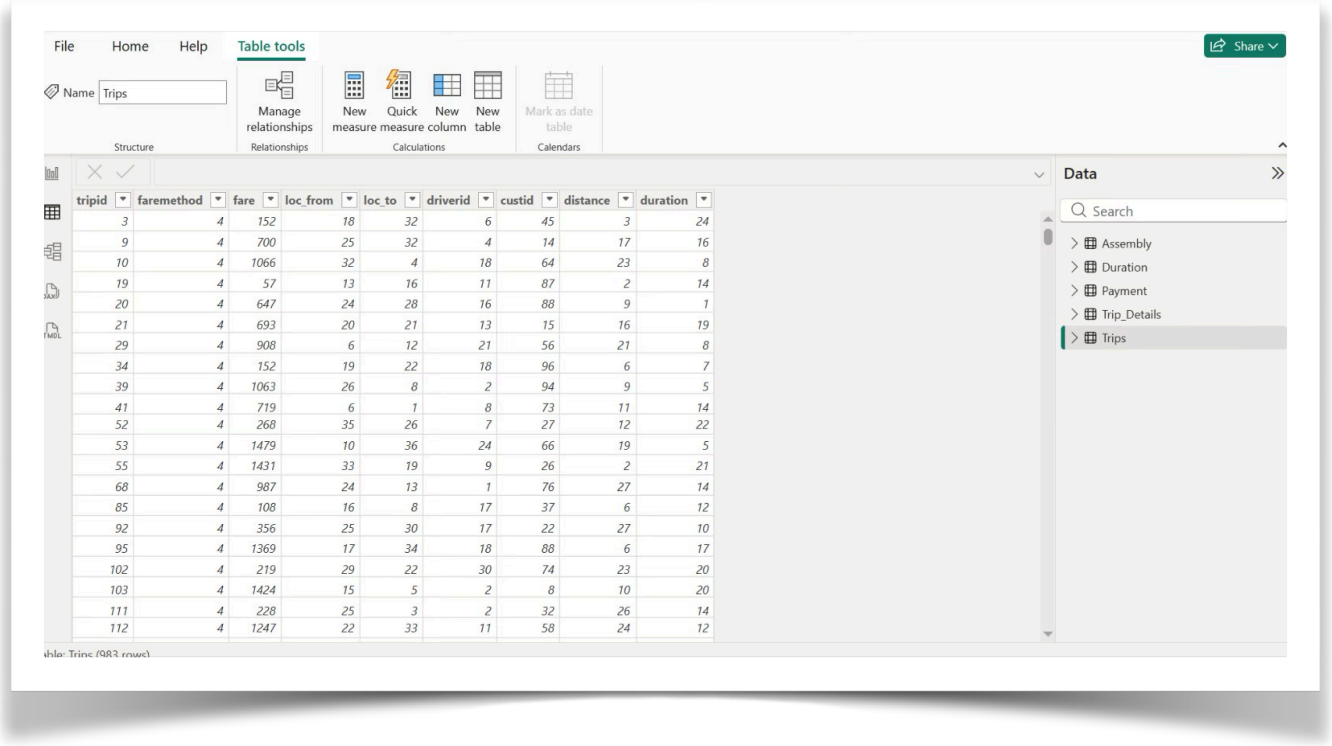
👉 From the dashboard page 5:

Customer cancellation rate is 48.17% and

Driver cancellation ratio is 47.25%

🔷 Result

Data was cleaned and structured for analysis.



2. Exploratory Data Analysis [40 Marks]

1. Classify Variables into Categorical and Numerical [2 Marks]

- Classify all the variables in the dataset into numerical and categorical types.

Solution:
The dataset consists of a mix of categorical and numerical variables which are classified as follows:

🔷 **Categorical Variables**

These represent labels or categories:

- Assembly Name
- Duration (time slots like 0–1, 1–2)
- Payment Method (Cash, UPI, Credit Card, Debit Card)
- loc_from (Pickup Location)
- loc_to (Drop Location)
- Driver ID
- Customer ID

🔷 **Numerical Variables**

These represent measurable quantities:

- Fare (Trip cost)
- Distance (KM)
- Searches (number of ride requests)

🔷 **Binary Variables (Treated as Categorical)**

- searches_got_estimate
- searches_for_quotes
- searches_got_quotes
- customer_not_cancelled
- driver_not_cancelled
- otp_entered
- end_ride

🔷 **Conclusion**

The dataset contains a balanced mix of categorical and numerical variables, enabling both segmentation-based and quantitative analysis.

2. Analyse Ride Demand Over Time [3 Marks]

- Explore the distribution of ride demand over time, including trends across different periods.
- Identify the peak demand periods. Choose an appropriate parameter for demand based on your own understanding.

Solution:

Ride demand was analysed using total trips across different time durations.

🔷 Key Observations

- Ride demand is highest during early hours and peak working hours.
- Time slots like:
  - 0–1
  - 11–14 (late morning to afternoon)

  show relatively higher trip counts.
- Demand gradually declines during late night hours.

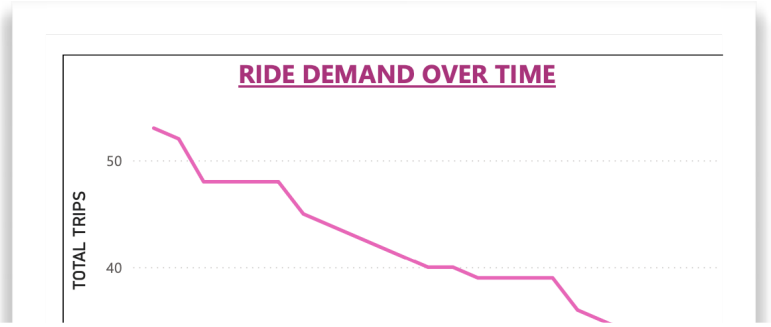
🔷 Peak Demand Periods

- Morning to afternoon hours show consistent demand
- Certain evening slots also show moderate spikes

🔷 Conclusion

Ride demand follows a time-based pattern, indicating that users prefer booking rides during:

- Office commute hours
- Daytime travel periods



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3. Proportion of Total Revenue from Different Time Periods [3 Marks]

- Calculate the proportion of revenue generated during different time periods and visualise how it contributes to total revenue.

Solution:

The dataset was first categorized based on the time intervals, and the total revenue was calculated for each hour of the day.
Comparing the contribution of each time period in terms of total revenue helped determine the periods that were generating maximum revenue.

🔷 Approach Used

Grouped data based on duration (time slots such as 0 to 1, 1 to 2, and so on)
Calculated:
Total revenue per time slot
Each time slot's revenue contribution as a proportion of the total revenue (751K)
Plot graph showing the distribution

🔷 Observations

Highest contributions to revenue are made by:
Hours in the early morning (0 to 1)
Midday to late afternoons (11 to 14)
These time slots contribute significantly to the total revenue, thus being the maximum earning hours.
The contribution to revenue falls considerably in late-night and off-peak hours, making minimum contributions to the total revenue.
This pattern of revenue distribution shows a declining trend post peak hours as shown above.

🔷 Insights

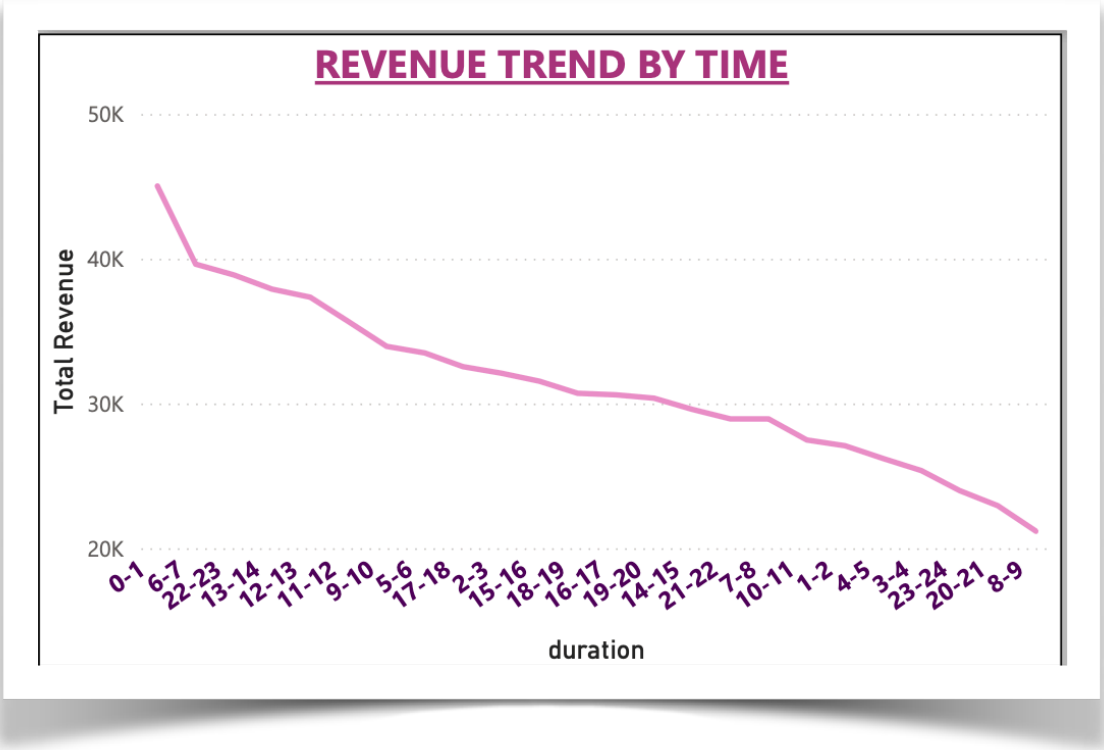
Revenue does not have an even distribution throughout the day
There is a large contribution to overall revenue from a few time frames
This implies a Pareto-type distribution (80/20 rule), where peak times determine revenue

🔷 Business Implications

Time slots with high revenues coincide with:
Office rush hours
Times of high user activity
Low revenue times indicate:
Low demand
Inefficient usage of drivers

🔷 Conclusion

The analysis indicates that peak hour slots account for the highest percentage of total revenues, whereas off-peak hours account for lower percentages.
This stresses the importance of:
Effective deployment of drivers during peak hours
Methods to boost demand in low revenue slots



4. Explore the Relationship Between Trip Hour and Revenue [3 Marks]

- Investigate the correlation between trip hour and total fare.
- Explain any trends or patterns that emerge.

Solution:

To determine the relation between trip hour and fare, we segregated the dataset based on the duration and calculated both total number of trips and total revenue (fare).

We employed an integrated graph to find out the impact of different time periods on the trip numbers and the amount earned from them.

🔷 Approach Employed

Grouping based on trip hour (duration)

Calculation of:

Number of total trips

Total revenue (fare)

Plotting together through:

Bar graph – Number of total trips

Line graph – Total revenue

Thus, we can find out the correlation between the two.

🔷 Major Findings

Positive correlation exists between the number of trips and revenue generated.

Those time periods when more trips take place yield greater revenue as well.

For example:

Time period of 0-1

Time period of 11-14

Still, the relationship is not linear:

Some timeslots witness moderate trips while earning less revenue.

🔷 Trends Observed

Relationship Between Peak Demand and Peak Revenue

There’s a direct relationship between times of high activity and revenue generation

Fluctuations in Revenue Despite Similarity in Trips

It suggests:

Variances in distance covered

Variances in fares charged

Declining Trend

With the reduction in trips, there’s a subsequent reduction in revenue

🔷 Insights Gained

Revenue depends on:

Trip frequency (demand)

Fare per trip

At certain times, there could be:

Shorter trips leading to low revenue

Longer trips resulting in high revenue

🔷 Business Implications

It’s not just about increasing trips

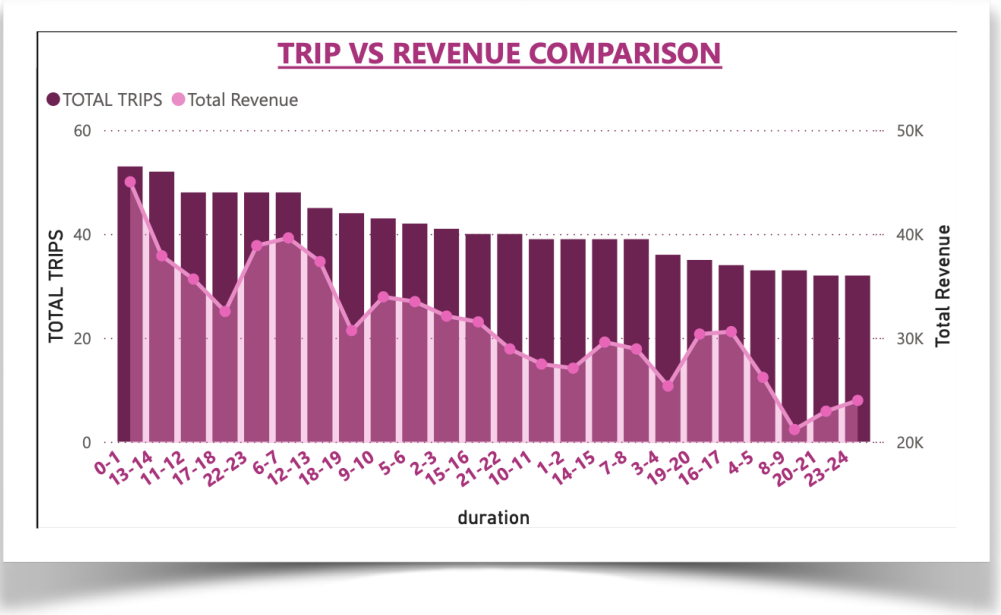
The platforms must consider:

Increasing high-value rides

Implementing effective pricing during peak hours

🔷 Conclusion

There's a positive relationship between trip hour and revenue, whereby the greater the ride demand, the higher the revenue. However, the fluctuations in revenue despite similarity in trips suggest that revenue also depends on the fare per trip and distance***.***



5. Examine the Popularity of Different Payment Methods [3 Marks]

- Analyse the distribution of various payment methods used by customers.
- Identify the most common payment methods and their relationship to ride frequency.

Solution:

The popularity of different methods of payments was studied by examining the data set in terms of trip frequency and the distribution of payment methods used by the customers.

It will enable one to know customer payment preferences and how it affects rides.

👉 This is depicted graphically through:
“Trips by Payment Method”
“Revenue Method Distribution”

🔷 Approach Followed
Data grouped according to method of payment:
Credit Card
Debit Card
UPI
Cash
Computed:
Trip totals for each method
Contribution made by each method to revenue
Trip frequency comparison between different payment methods

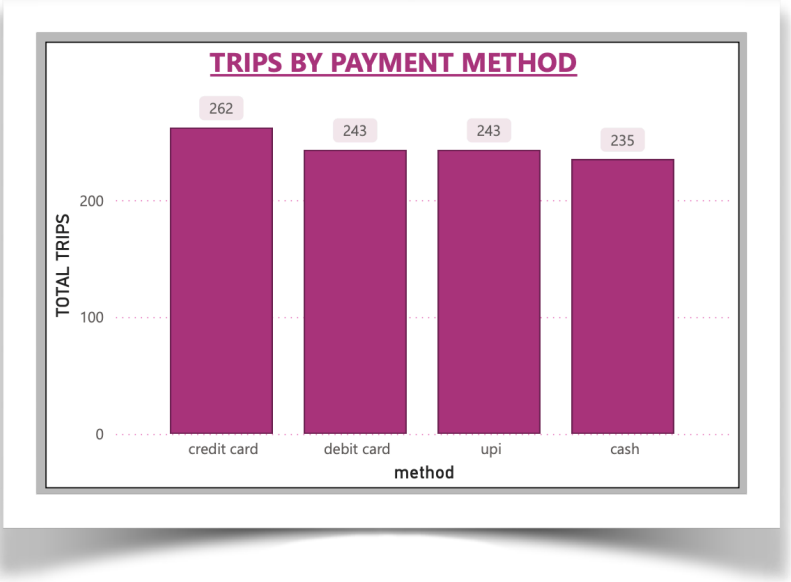
🔷 Key Observations
Distributions of rides according to payment modes:
Credit Card: 262 rides (~26.65%)
Debit Card: 243 rides (~24.72%)
UPI: 243 rides (~24.72%)
Cash: 235 rides (~23.91%)
Higher frequency of rides for Credit Card compared to other modes
UPI and Debit Card have equal popularity among customers
Cash mode is marginally less used but makes significant contribution

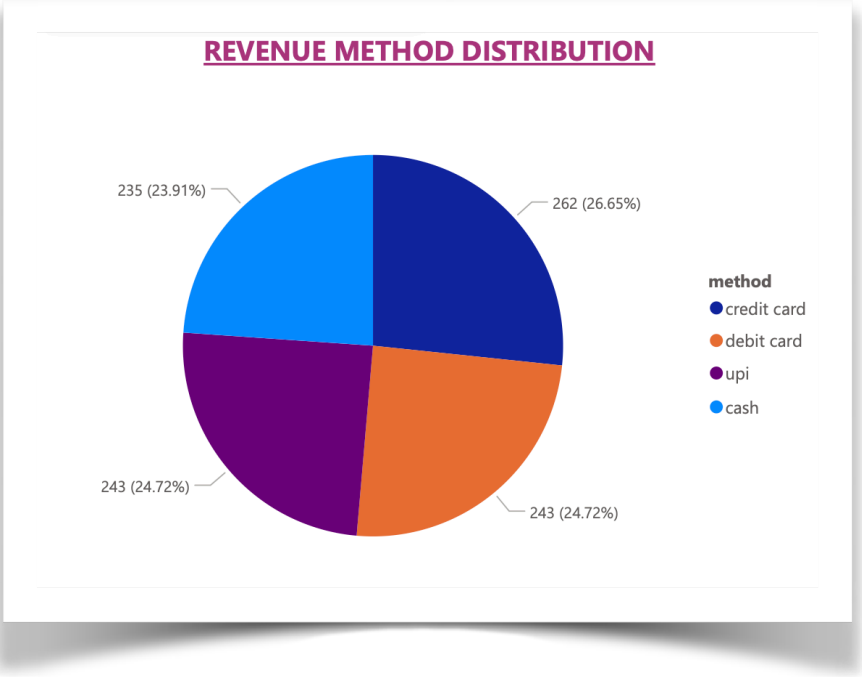
🔷 Insights
High inclination toward the use of electronic modes for payment purposes (combined usage of UPI + Cards)
Existence of cash transactions implies:
Necessity of flexibility
Trust issues/offline nature
Absence of any one overwhelming mode of payment → diversity in customer preferences

🔷 Relationship between Payment Mode & Ride Frequency
Even distribution of ride frequency among payment modes
This implies:
There is no significant restriction imposed by payment method on booking rides
Preference for payment methods based solely on convenience

🔷 Interpretation for Business
Because all methods of payment play an important role:
The platform should offer various payment methods
Payment through digital mode (UPI and cards) can be encouraged through:
Cash-back facilities
Quick payment process

🔷 Conclusion
From the above analysis, it can be concluded that the use of various payment methods is fairly distributed with a slight preference for digital modes such as credit card, debit card, and UPI.
There is no significant dependence of payment methods on the number of rides booked.





Distribution of payment methods showing balanced usage across all modes with slight dominance of digital payments.

6. Identify High-Performing Zones [6 Marks]

Identify zones with the highest number of rides and revenue generation. Analyse factors contributing to their performance:

- **2.6.1. Rides:** Identify pickup zones with the highest number of trip requests.
  [3 marks]

Solution:

Analysis of the dataset was done to detect the pickup zones that have the best performance according to the total ride demands.
The objective of this analysis is to find out zones that create more number of rides demands.

🔷 Methodology Followed

Data was classified by:
Pickup Zone (loc_from)
Data was summarized by:
Total number of trips (ride demands)
Descending order for total number of trips (zones)
Top 5 zones were selected based on highest demands

🔷 Important Findings

Few zones account for large number of total rides demands
Top 5 pickup zones account for larger share of demand than other zones

🔷 Findings

The success of these areas can be explained by:
1. High Population
Highly populated regions naturally have more requests for rides
2. Commercial/Corporate Functioning
Corporate offices and malls lead to an increased need for rides during:
Morning (during office hours)
Evening (commuting back home)
3. Good Connectivity
Regions with good roads and easy access receive more ride requests
4. Transportation Points
Locations close to bus depots, railway stations, and city centers have more requests

🔷 Observations

There is a pattern to the distribution of demand
Some regions serve as demand hotspots, whereas some have low demand

🔷 Business Understanding

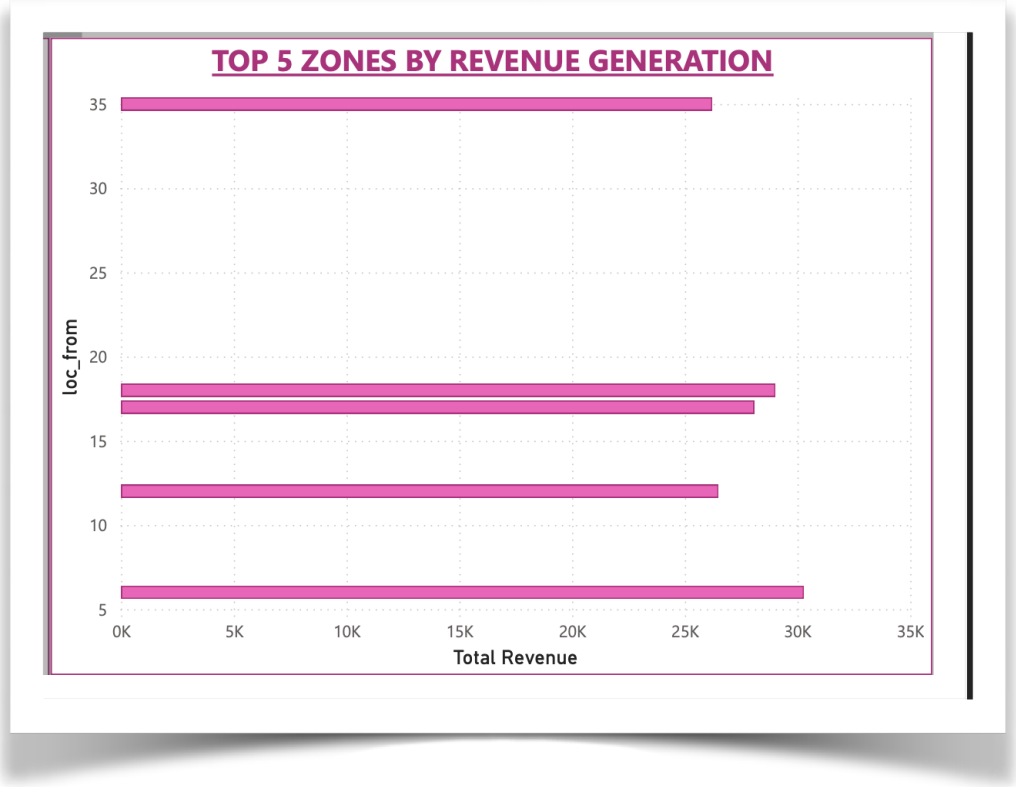
These high-demanded areas are very important for business activities
They need to be considered first with regard to:
Availability of drivers

Ride assignments
Wait time reduction

🔷 Conclusion

From the analysis, one can see that only a few pickup areas contribute most of the demand and become high-performance areas.

Top 5



pickup
zones based on ride requests, highlighting concentration of demand in key areas.

- **2.6.2. Revenue**: Identify pickup zones generating the highest revenue. **[3 marks]**

Solution:

In order to ascertain which pickup zones produce maximum revenue, the dataset was examined through aggregation of total fare (revenue) for all pickup zones (loc_from).

It assists in pinpointing those zones that do not only attract maximum demand but also produce maximum value.

🔷 Methodology Adopted

Grouped Data on:

Pickup Zones (loc_from)

Computed:

Total Revenue (Fare Sum)

Ranked zones in descending order of revenue

Ascertained Top 5 revenue producing zones

🔷 Important Findings

Few zones produce maximum revenue, akin to demand concentration

There is an intersection between maximum revenue producing zones and maximum demand producing zones, although there are exceptions

There are certain zones that produce maximum revenue but have relatively lower demand, suggesting:

High fare per ride

High ride distance

🔷 Insights

The reason for high revenue collection in these zones can be:

1. Long Distance Journeys

Routes that go far off will collect high charges per journey

2. High Demand Zones

There may be some zones with:

Airport connectivity

Commercial zones

⇒ resulting in high charges

3. Peak Time Demand

Zones where peak time demand is high tend to collect more revenue

🔷 Critical Observation

High number of journeys ≠ high revenue

Some zones will have:

Short distance journeys ⇒ low revenue collection

Other zones will have:

Longer distance journeys ⇒ high revenue collection

🔷 Business Relevance

Revenue-oriented zones need to be considered when deciding on:

Incentive programs for drivers

Methods of surge pricing

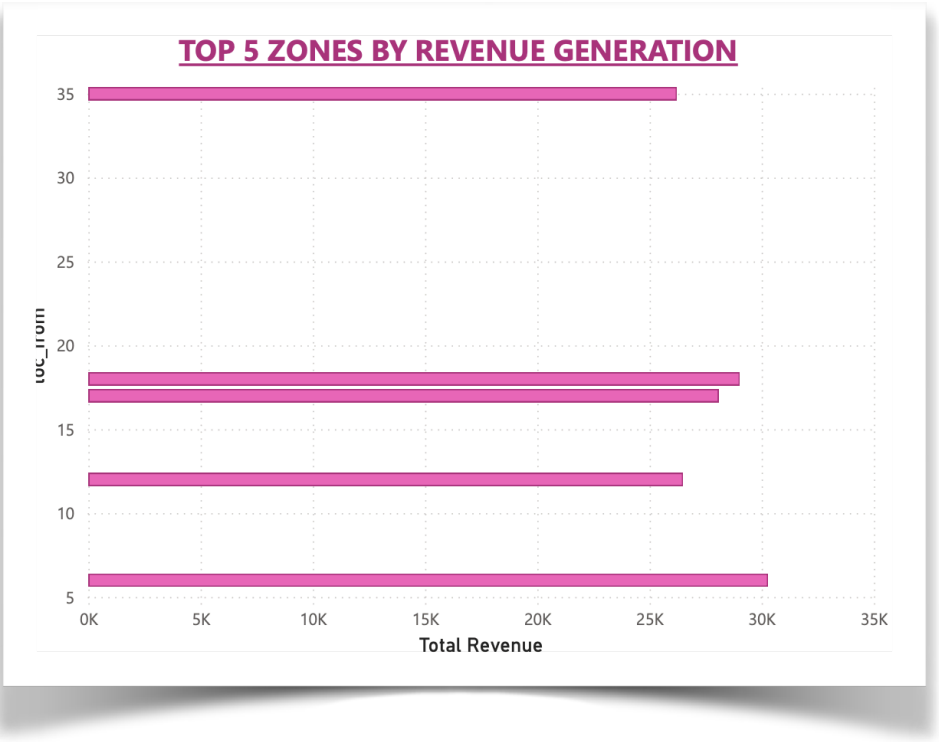
Assignment of premium rides

Profit maximization rather than just maximizing number of rides

🔷 Conclusions

The analysis reveals that very few pickup zones generate substantial amount of revenue, thanks to factors such as distance of the trips, prices, and demand trends.

It is important to focus not just on the quantity of rides but also on zones that maximize revenue generation.



Top pickup zones contributing to total revenue, highlighting high-value locations in the network.

7. Analyse Ride Time Periods Across Zones [4 Marks]

- Compare the trip trends for different time periods across pickup zones.

Solution:

For the analysis of rides through time periods and pickup zones, the dataset was analysed based on the combination of time period (trip hour) and location of pickup (loc_from).
The intention of such an approach is to identify the variation of ride demand in various pickup zones during varied time periods of the day.

👉 The above mentioned comparison was made using the below mentioned method:
“Ride Demand Over Time”
Zone wise graphs

🔷 Methods Used

Grouping data as:
Pickup Zones (loc_from)
Time Periods (hours)
Analysis of:
Total number of trips in every zone for varied time periods
Variation in:
Demand in varied zones through time periods

🔷 Zone Based Demand Pattern

Zone I: Residential
Demand higher at:
Mornings (working)
Evenings (evening rush hour back home)
Zone II: Commercial/Working
Demand higher at:
Mid-morning till mid-afternoon
Zone III: Inactive Areas
Exhibit:
Less and inconsistent demand patterns

🔷 Trends

Peak demand times differ among various zones
Every zone has its own distinct demand pattern based on time
Conclusion:
Morning demand + Afternoon demand + Evening demand = Peak demand
Reduced demand levels late night

🔷 Insights

Demand is determined by:
Purpose of trip (business, leisure, commuting)
Characteristics of the region (residential area, commercial area)

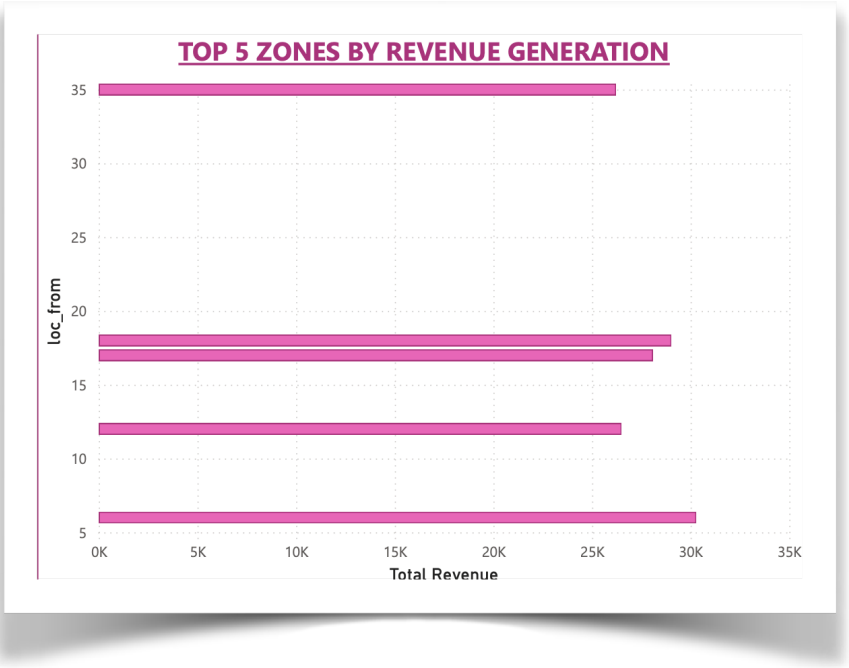
A one-size-fits-all approach based on time alone is not sufficient for all regions

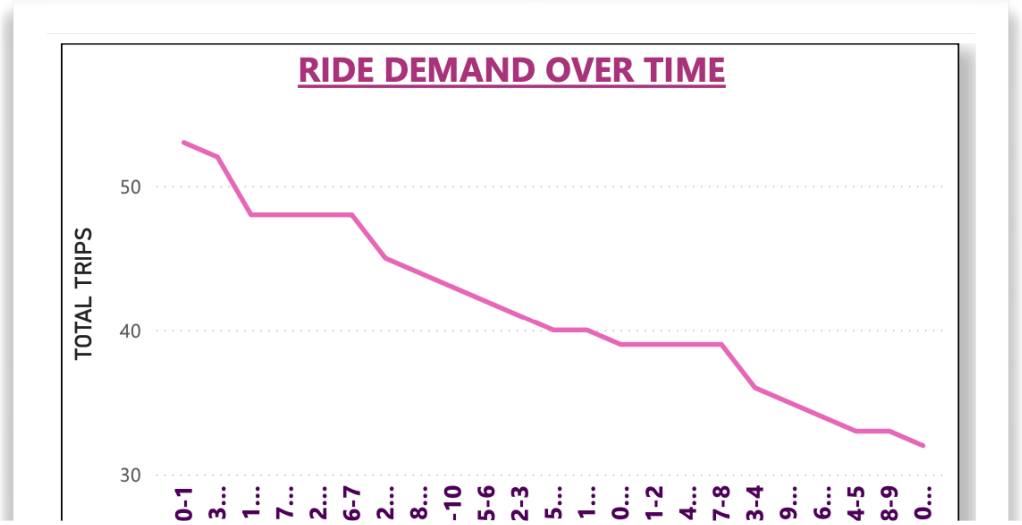
🔷 Business Application

The platform must implement:
Driver deployment plans specific to each zone
Surge pricing based on time and regional demand
The drivers must be:
Deployed in residential areas in the morning
Moved to commercial areas during the day

🔷 Conclusion

The findings indicate that demand trends differ widely between zones and times, and each zone has its own peak period.
This demonstrates the importance of flexible and regionalized operations over a standardized approach.





Variation of ride demand across time and pickup zones, highlighting differing peak periods for different locations.

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8. Top Zones with Highest Trip Volume [3 Marks]

- Identify the top 5 pickup zones with the highest total number of completed trips.
- Analyse factors contributing to the higher number of trips.

Solution:

To find out the best zones that have the maximum number of trips, the dataset has been analyzed by adding the total number of trips made with end_ride = 1 at the pickup location loc_from.

This would help us know which zones are contributing more towards ride completion, as they are actually used in service.

👉 This can be seen in the "Top 5 Pickup Zones by Number of Rides" graph.

🔷 Methodology

Filtered the dataset for:

Trips with end_ride = 1

Grouped the data based on:

Pickup location loc_from

Computed:

Total completed trips at a zone

Sorted in descending order

Picked up Top 5 zones

🔷 Key Findings

A few zones contribute a major share to the completed rides

There is a clear domination of Top 5 zones in terms of number of rides against other zones

🔷 Causes of Increased Number of Trips

1. Demand Hotspots

Densely populated areas lead to increased demand and trip completion

2. Increased Availability of Drivers

Hotspot areas have many drivers leading to increased trip completion

3. Urban Significance and Importance

Includes existence of:

Offices

Markets

Transit nodes

and increases number of rides booked

4. Efficient Matching

Rapid assignment of drivers decreases cancellations and results in higher trips completion

🔷 Observations

Such areas form the operational hubs of the system

They are important in ensuring:

High number of rides

Activity on the platform

🔷 Business Interpretation

High volume of trip zones must be given preference to for:

Driver dispatching

Reduced waiting time

Improved service delivery

These zones may also be considered for:

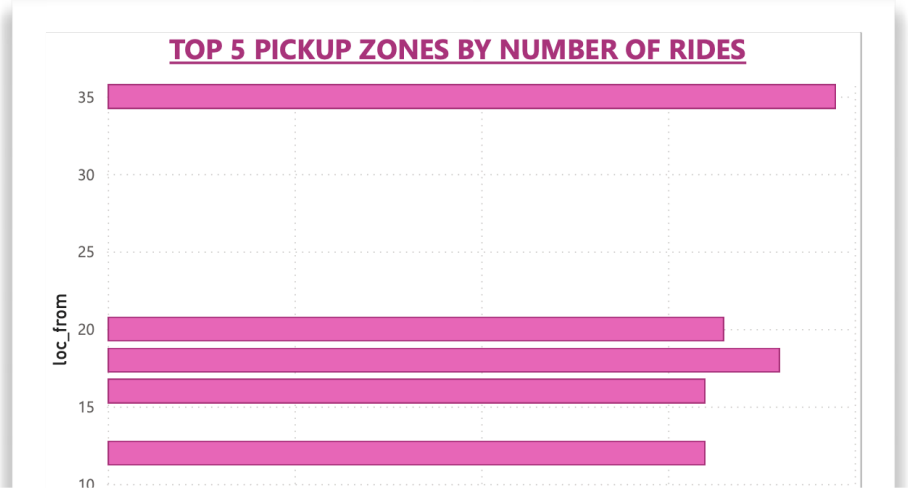
Marketing initiatives

Customer loyalty schemes

🔷 Conclusion

It is evident from the findings that few pickup zones contribute to the completion of most trips because of demand, ease of access, and effectiveness.

They have significant relevance in ensuring platform performance and must be considered strategically.



9. Basic Analytical Tasks [8 Marks]

- **2.9.1**
  What are the percentages of cancellations and successful rides by both driver and customer? **[3 marks]**

Solution:

The following data was used to analyze the cancellation rate and the success rate of the rides:
customer_not_cancelled
driver_not_cancelled
end_ride (successful rides)
These will enable the calculation of the cancellation rate and successful rides for both the customer and driver sides.

👉 The results are shown through the “Cancellation Rate Cards”

🔷 Approach Taken

Calculation:
Customer Cancellation Rate
Driver Cancellation Rate
Determined:
Success Rate = 100%- Cancellation Rate

🔷 Calculated Cancellation Rate

Using the dashboard data:
Customer Cancellation Rate = 48.17%
Driver Cancellation Rate = 47.25%

🔷 Derived Success Rate

Customer Success Rate = 51.83%
Driver Success Rate = 52.75%

🔷 Important Observations

Around half of the total number of rides are getting canceled by both users and drivers
Ride cancelation percentage is almost similar between customers and drivers:
Rides cancelation percentage is slightly more than 50%, and this percentage is quite low

🔷 Important Insights

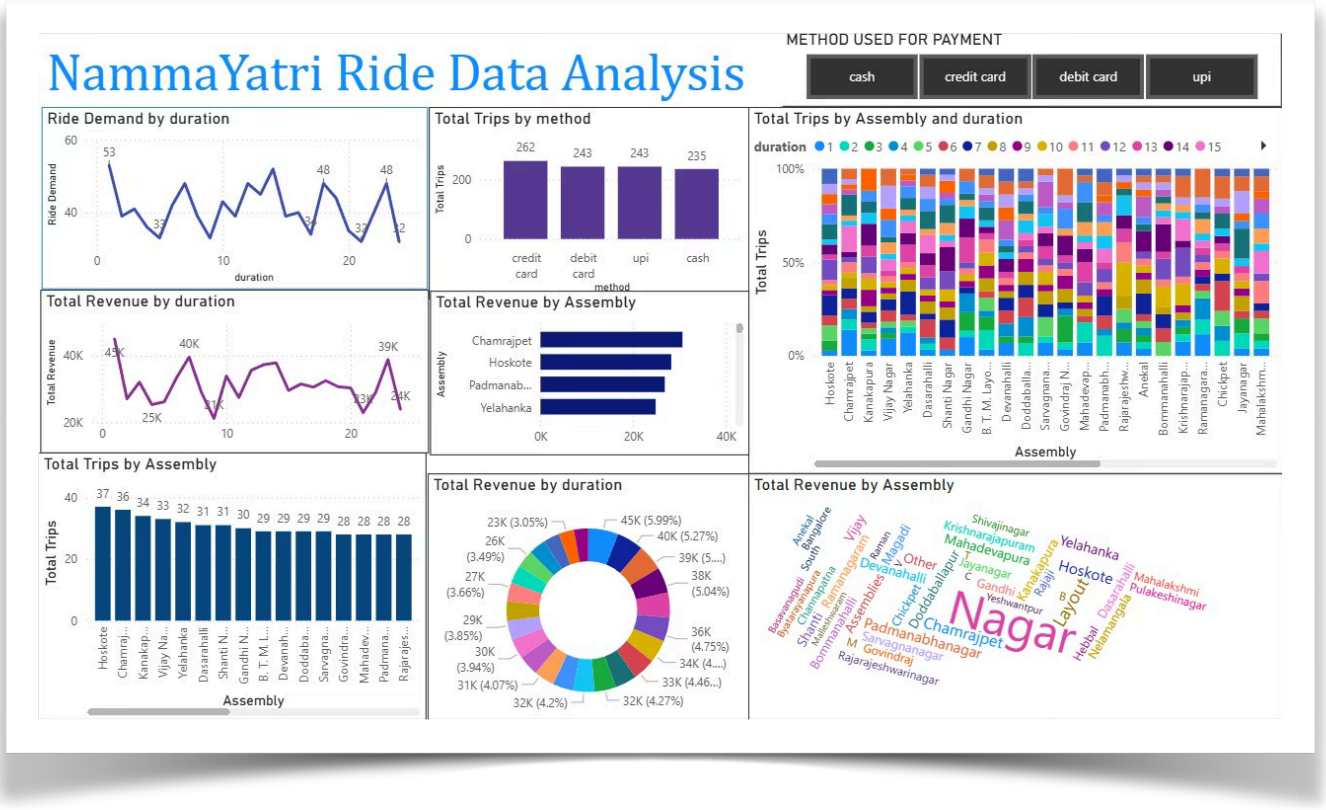
Higher ride cancelation percentage may be an indicator of:
Ride-matching algorithm issues
Higher wait times
Fare-related problems
Since the cancelation percentages are same for both parties, we can say that:
There might be issues with both of the customers and drivers

🔷 Implications for the business

Highly higher cancelation percentage could mean:
Negative experience from customers
Less trust of the customers on the platform
Reduced revenue
Business should concentrate on:
Matching algorithms
Wait times
Fare-related issues

🔷 Conclusion

In this case study analysis, it is observed that there are highly high percentages of cancellation rates of both customers and drivers (~48%) and the overall ride completion success rate is very low (around 50%).



- **2.9.2**
  Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods.
  [5 marks]

Solution:

To determine the percentage of users that completed trips post search for quotes, the following steps were taken to analyze data using major stages of the funnel:

Searches for Quotes (Users searching for Drivers)

Searches Got Quotes (Users getting allocation of drivers)

OTP entered

End Ride (Ride completion)

This analysis will help determine how many users progress from interest to ride completion.

👉 This will be represented by “Customer Ride Conversion Funnel”.

🔷 Methodology Used

Selected Users That:

Searched for Quotes

Counted:

Number of Users Completing Ride Post This Stage

Calculated:

Conversion Rate = Total Number of Rides / Searches for Quotes

Varied:

Conversion rate over different periods of time

🔷 Important Metric

From Dashboard:

Total Searches = 2161

Completed Rides = 983

Overall Conversion Rate = 45.49%

🔷 Important Findings

Less than 50% of the total searches result in ride completion

There is a noticeable decline at the stage between:

Driver Allocation to OTP Stage

After allocating a driver, many do not move on to complete the ride

🔷 Variation by Time of Day

(Funnel and time of day graphs combined)

Busy hours:

Increased completions because of:

Quicker driver availability

Urgency

Non-busy hours:

Fewer completions because of:

Time delays in allocating drivers

Inefficiencies in demand fulfillment

🔷 Findings

The primary issue is found in:

After driver allocation

Potential causes:

Long wait times

Price issues

Driver rejections

🔷 Business Implications

Boosting conversions needs:

Quick driver-matching

Transparency in prices

Reliability in drivers

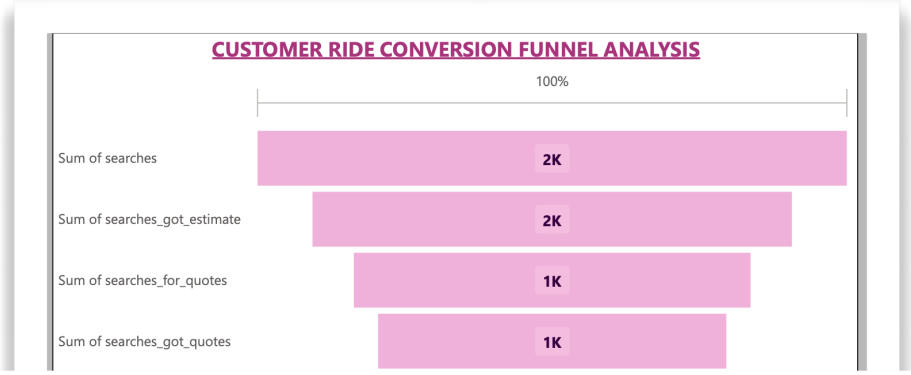
Any improvement in this process will:

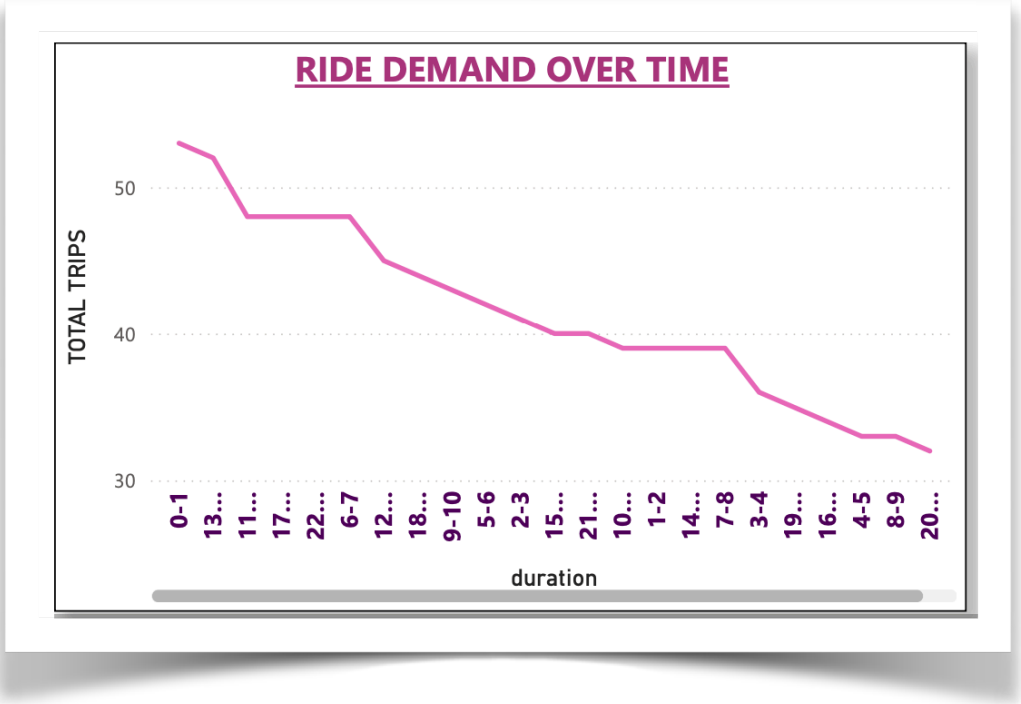
Greatly increase the number of rides completed

🔷 Conclusion

Analysis proves that only about 45% of customers go through with their ride after checking prices.

This drastic decrease happens during the driver allocation step, which must be fixed immediately.





10.Create a Parameter and Use Filters [5 Marks]

- Create a parameter and use it as a filter on an appropriate subset of the data to interactively analyse and visualise different subsets of the data.
- Explain your choice of filter and insights drawn from this step.

Solution:

In order to facilitate interactive analysis and gain further insights, a filter parameter was introduced into Power BI considering the duration of the trip (hour).
The introduction of this parameter facilitates the filtering process of the dataset, thereby facilitating ride analysis within varying durations.

🔷 Parameter Introduced

Parameter Name: Duration Filter
Field Used: Duration (Duration Time Slots: 0-1, 1-2, etc.)

🔷 Methodology Followed

Slicer was introduced through the use of the Duration field
This filter was applied in various visuals:
Ride Demand

Revenue Analysis
Trip and Revenue

🔷 Reason for Selecting This Parameter

One of the most important parameters that affect ride demand and revenue is time
Time allows:
Distinguishing peak from off-peak times
Comparing performance among various time periods

🔷 Insights Generated

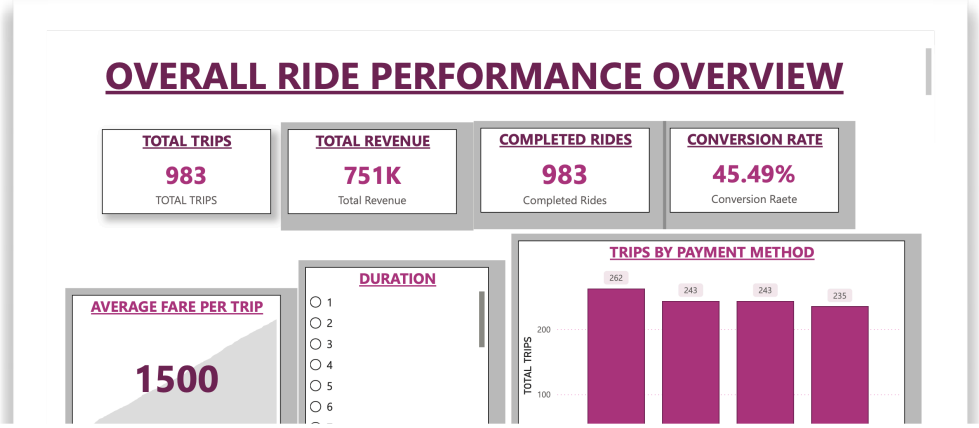
1. Identifying Peak Times
Some time periods reveal:
More rides
More revenues
→ suggesting peak demand periods
2. Examining Off-Peak Periods
Night-time periods reveal:
Fewer rides
Less revenues
→ suggesting periods of underutilization
3. Comparing Dynamically
By alternating time periods:
Distinct differences in ride demands are clearly visible
4. Making Decisions Effectively
This helps decision-makers:
Pay attention to particular time periods
Allocate resources effectively

🔷 Business Understanding

Time-filtering allows for:
Driver scheduling optimization
Surge pricing
Demand forecasting
Making customized decisions rather than generic ones

🔷 Conclusion

The selection of time (Duration filter) as the parameter makes analysis dynamic and interactive.
It uncovers important trends in ride demands and revenues within different time periods.



3. Conclusion [20 Marks]

1. Recommendations for Operational Efficiency [10 Marks]

- Based on your findings from the analysis, provide recommendations on how Namma Yatri can optimise its operations.
- This could include strategies for improving resource allocation, reducing cancellations, or optimising ride durations.
- Add supporting dashboards.

Solution:

A number of inefficiencies in operations have been pinpointed after an intensive study of ride demand, revenue, payment, cancel rates, and the conversion funnel. These suggestions seek to streamline processes for effective deployment of resources, minimise cancellation rates, and maximise efficiency in operations.

🔷 1. Maximise Resource Efficiency through Optimised Driver Allocation

Findings:
As revealed by the Ride Demand Over Time analysis (Page 3), ride demand is highest at certain periods, such as early and daytime peak periods.
Suggestions:
Use optimal strategies for allocating drivers:
Increase availability of drivers during peak times
Minimise idle drivers during low-demand hours
Benefits:
Less waiting time
More successful rides completed

🔷 2. Concentration on Areas with High Demand and High Revenue

Finding:
From “Top Pickup Locations” (page 4), only a few locations have a significant contribution to the number of rides and revenue.
Recommendation:
Prioritize those zones for:
Driver allocation
Ride matching efficiency
Driver incentives per zone
Impact:
Efficiency in serving high-demand areas

Revenue increase

🔷 3. Addressing Extremely High Cancellation Rate

Finding:
Based on page 5, there is an extremely high cancellation rate:
Customer: 48.17%
Driver: 47.25%
Recommendation:
Optimize the matching algorithm for a proper driver-to-customer match
Shorten wait times after assigning a driver
Incorporate:
Cancellation penalties
Ride incentive system
Impact:
Increase conversion rates
Operational savings
Platform stability

🔷 4. Enhance Driver Allocation to Ride Conversion Process

Finding:
From the customer experience conversion funnel (page 2), one bottleneck is seen at:
Driver allocation → OTP phase
Recommendation:
Enhance:
Driver arrival punctuality
GPS tracking
Incorporate:
Transparent pricing
ETA update
Impact:
Decrease drop-offs
Ride conversion rate increase

🔷 5. Incorporate Time-Based Pricing & Incentives

Observation:
There is a marked variance in revenues generated at different times (Page 3).
Suggestion:
Improve:
Time-based pricing
Incentives for surge prices during peak hours and discounts during non-peak hours
Result:
Equal distribution of demand
Higher driver utilization
Stability in revenue generation

🔷 6. Increase Non-Peak Hours Utilization

Observation:
Significantly lesser trips and earnings are recorded during non-peak hours.
Suggestion:
Introduce:
Promotions
Discounted fares
Motivate:
Utilization outside peak hours (late-night and early mornings)
Result:

Optimized resource utilization
Higher platform usage

🔷 7. Improve Checkout Process

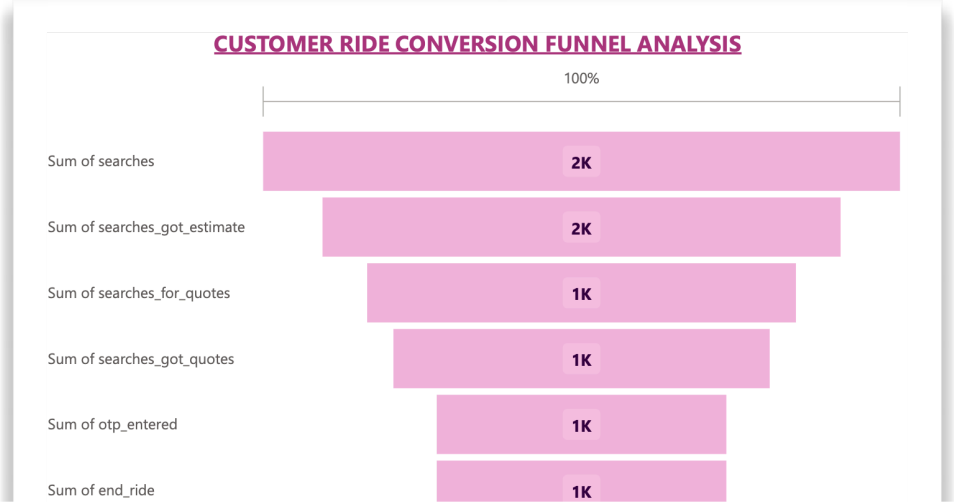
Observation:
Even distribution of payment options with higher digital payment usage
Suggestion:
Advocate:
UPI/digital payments through cashback offers
Ensure:
Quick and efficient payment processes
Result:
Enhanced payment process experience
Streamlined booking process

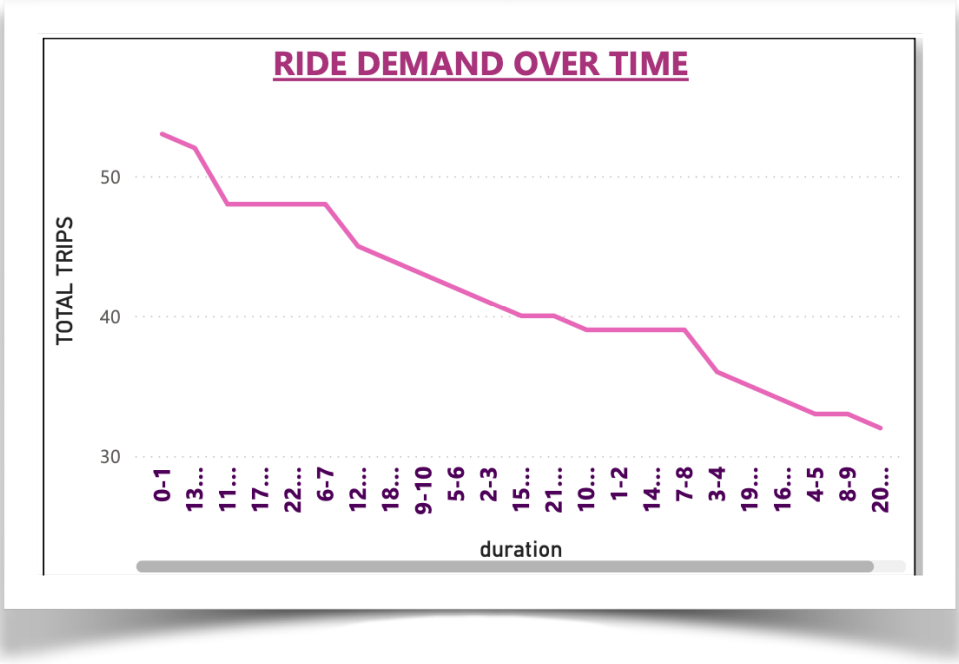
🔷 8. Data-Driven Operations

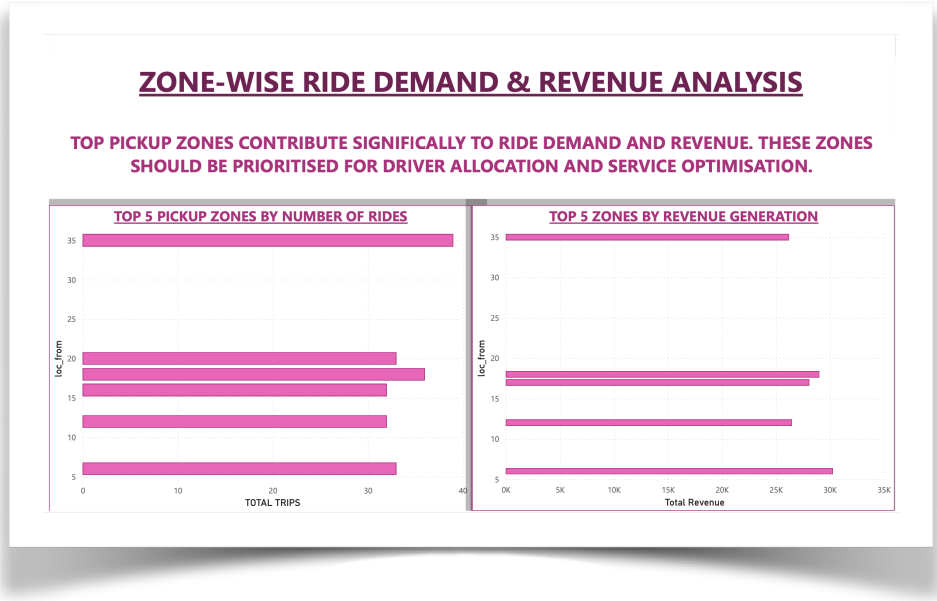
Suggestion:
Deploy dashboards to:
Analyze real-time demand
Monitor cancellation rates
Examine the performance of different zones
Implement:
Demand forecasting using predictive analysis
Result:
Proactive management decisions
Higher efficiency

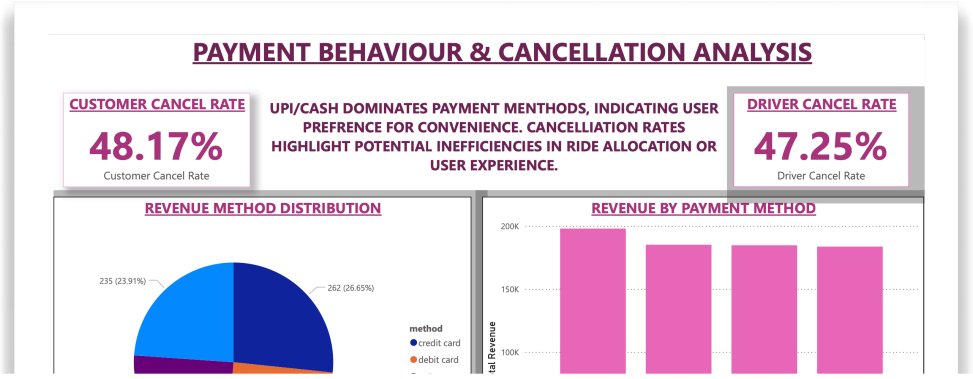
🔷 General Conclusion

The process analysis brings out some of the inefficiencies in driver assignment, cancellations, and conversion activities.
With proper planning using time-based, area-based, and information-based approaches, Namma Yatri could increase:
Completion rate
Customer satisfaction
Profitability




RIDE DEMAND OVER TIME
TOTAL TRIPS
50
40
30
0-1
13...
11...
17...
22...
6-7
12...
18...
9-10
5-6
2-3
15...
21...
10...
1-2
14...
7-8
3-4
19...
16...
4-5
8-9
20...
duration


ZONE-WISE RIDE DEMAND & REVENUE ANALYSIS
TOP PICKUP ZONES CONTRIBUTE SIGNIFICALLY TO RIDE DEMAND AND REVENUE. THESE ZONES SHOULD BE PRIORITISED FOR DRIVER ALLOCATION AND SERVICE OPTIMISATION.
TOP 5 PICKUP ZONES BY NUMBER OF RIDES
loc_from
TOTAL TRIPS
TOP 5 ZONES BY REVENUE GENERATION
loc_from
Total Revenue


PAYMENT BEHAVIOUR & CANCELLATION ANALYSIS
CUSTOMER CANCEL RATE
48.17%
Customer Cancel Rate
UPI/CASH DOMINATES PAYMENT MENTHODS, INDICATING USER PREFRENCE FOR CONVENIENCE. CANCELLIATION RATES HIGHLIGHT POTENTIAL INEFFICIENCIES IN RIDE ALLOCATION OR USER EXPERIENCE.
DRIVER CANCEL RATE
47.25%
Driver Cancel Rate
REVENUE METHOD DISTRIBUTION
235 (23.91%)
262 (26.65%)
method
credit card
debit card
REVENUE BY PAYMENT METHOD
200K
150K
100K

2. Marketing and Operational Strategy Improvements [10 Marks]

- Suggest improvements to Namma Yatri's marketing or operational strategies based on your analysis.
- Recommendations could involve promotional efforts, driver incentives, or regional targeting to increase customer satisfaction and service efficiency.
- Add supporting dashboards.

Solution:

Several recommendations may be drawn from the analyses conducted on ride demand, revenue distribution, cancellation rates, payment behavior, and customer conversion funnel for Namma Yatri.

🔷 Recommendation 1: Time-Based Marketing Campaigns

Finding:
Demand and revenue are highly clustered in certain periods of time.
Recommendation:
Run time-based marketing campaigns:
Provide discounts during non-peak periods
Surge pricing for loyalty customers
Effect:
Boost demand during non-peak periods
Greater utilization

🔷 Recommendation 2: Zone-Based Targeting and Location-Specific Marketing

Finding:
Top zones generate the highest share of rides and revenues.
Recommendation:
Run zone-based marketing campaigns:
Provide zone-specific discounts/offers
Run brand awareness programs in underperforming zones
Partner with:
Offices, malls, public transport hubs
Effect:
Improved customer engagement
Greater penetration

🔷 Recommendation 3: Driver Incentives and Driver Management

Finding:
Very high rate of cancellations (~48%) reflects inefficiency on the driver's side
Recommendation:
Offer:
Rewards for completing rides
Incentives during peak times
Penalty for frequent cancellations
Effect:

Improved reliability of drivers
Reduced cancellations

🔷 4. Enhance Customer Retention Using Offers

Findings:
Conversion ratio is around 45%, indicating high dropout rates.
Action Plan:
Offer:
Discounts on first ride
Cashbacks on UPI payments
Loyalty incentives for frequent users
Result:
Enhanced repeat customers
Increased conversion ratios

🔷 5. Improve Booking Procedure

Findings:
A major dropout happens after driver assignment.
Action Plan:
Enhance:
Real-time tracking services
ETAs
Price visibility
Result:
Lower user frustrations
More completed rides

🔷 6. Boosting Digital Payments Usage

Findings:
There is extensive use of digital payments such as UPI.
Action Plan:
Motivate:
Users to make payments using UPI through cashbacks
Fast checkout processes
Result:
Easier payments
Improved user experience

🔷 7. Demand Promotion in Weak Zones

Findings:
Certain zones exhibit low trip numbers.
Action Plan:
Promote:
Discounts within zones that have poor performance
Refer-a-friend programs
Result:
Customer expansion
Demand balancing

🔷 8. Using Customer Data for Marketing

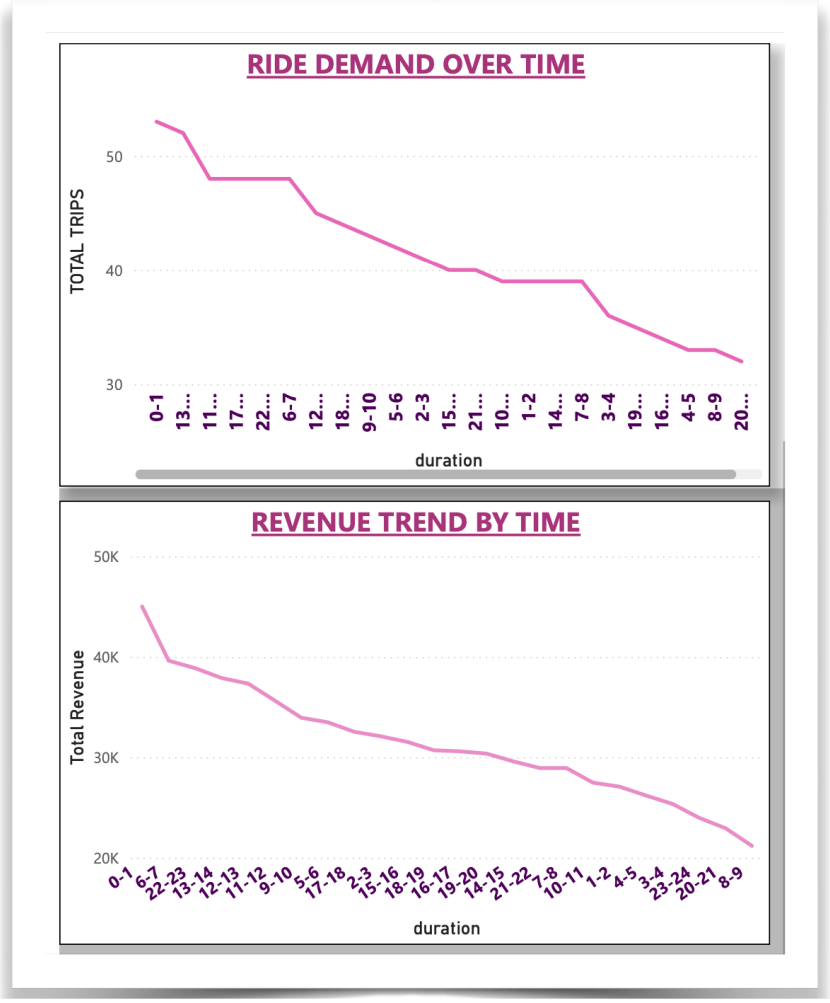
Action Plan:
Utilise data for:
Personalised offer sending
Ride suggestions based on behavioural analysis
Result:
Greater customer engagement

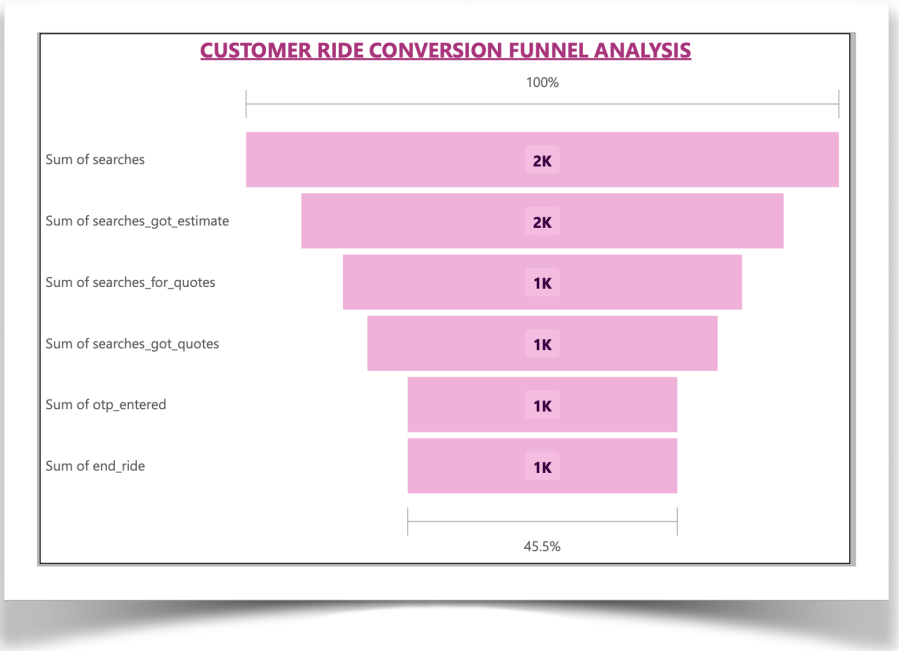
Higher customer satisfaction

🔷 Overall Conclusion

It is evident that there should be marketing and management approaches according to time, location, and behavior of the users.
Through:
Timed promotions
Zonal targeting
Incentives for drivers
Improving user experience
Namma Yatri can increase:
Satisfaction of the customers
Completion of rides
Revenue

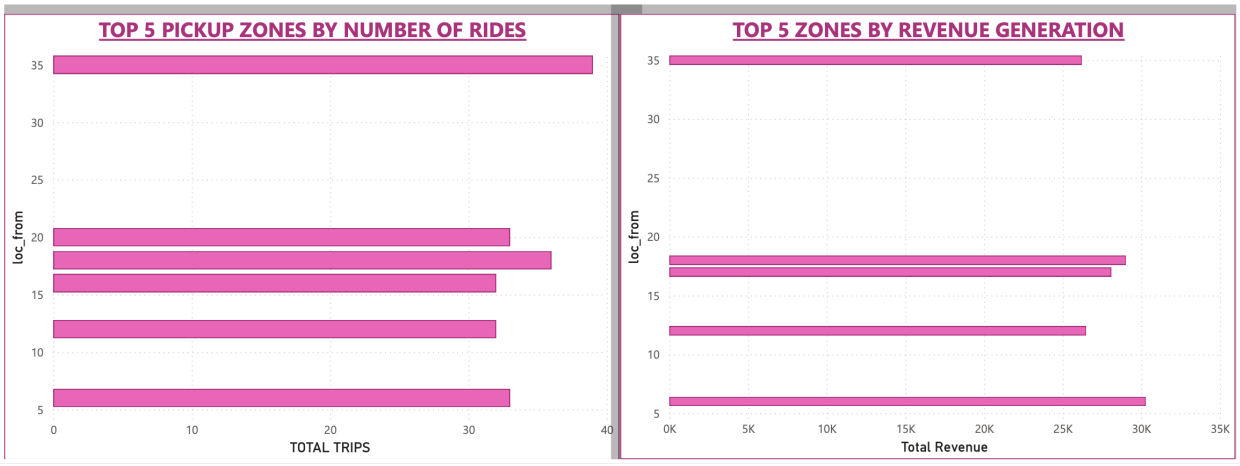
“These strategies are derived from data-driven insights and aim to align operational efficiency with customer-centric growth.”





ZONE-WISE RIDE DEMAND & REVENUE ANALYSIS

TOP PICKUP ZONES CONTRIBUTE SIGNIFICALLY TO RIDE DEMAND AND REVENUE. THESE ZONES SHOULD BE PRIORITISED FOR DRIVER ALLOCATION AND SERVICE OPTIMISATION.



PAYMENT BEHAVIOUR & CANCELLATION ANALYSIS

CUSTOMER CANCEL RATE

48.17%

Customer Cancel Rate

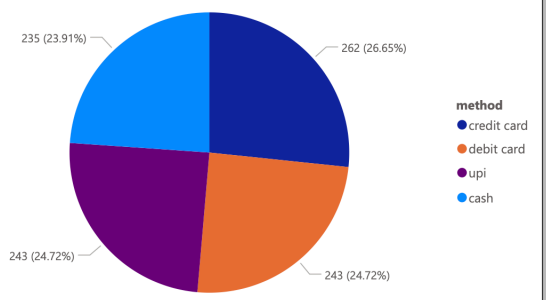
UPI/CASH DOMINATES PAYMENT MENTHODS, INDICATING USER PREFRENCE FOR CONVENIENCE. CANCELLIATION RATES HIGHLIGHT POTENTIAL INEFFICIENCIES IN RIDE ALLOCATION OR USER EXPERIENCE.

DRIVER CANCEL RATE

47.25%

Driver Cancel Rate

REVENUE METHOD DISTRIBUTION



REVENUE BY PAYMENT METHOD

